

Equations of Motion(EOM) Derivation

Modelling energy loss in Shock Absorber

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Introduction

Shock absorbers are critical components in vehicle suspension systems, designed to dissipate kinetic energy and reduce the impact of road irregularities. Proper modeling of a shock absorber allows engineers to analyze its damping characteristics, stiffness, and response to transient loads.

This work derives the equations of motion (EOM) for a simplified shock absorber model using the Lagrangian method. The resulting model is essential for simulation, ride comfort analysis, stability assessment, and suspension control system design.

System Description

The shock absorber is modelled as a one-degree-of-freedom (1-DOF) spring-damper-mass system.

Components:

- **Mass (m):** Represents the unsprung mass (wheel, axle, etc.).
- **Spring (k):** Represents the stiffness of the shock absorber.
- **Damper (c):** Represents the damping coefficient (viscous friction).
- **Base:** Connected to the vehicle chassis (moving or fixed).

The system is assumed to operate vertically.

Modelling Assumptions

1. The mass moves only in the vertical direction (1-DOF).
2. The spring is linear (force proportional to displacement).
3. The damper is linear viscous (force proportional to velocity).
4. The shock absorber housing is rigid.
5. Air resistance is neglected.
6. No internal friction other than the damper.
7. The base excitation (from road) is known or assumed sinusoidal.
8. These assumptions yield a linear second-order ODE.

Coordinate System and Generalized Coordinates

A fixed inertial frame is attached to the chassis or ground.

Generalized coordinate:

$$q = z$$

Where, z = vertical displacement of the mass relative to equilibrium.

Choice of Dynamic Modeling Method

The Lagrangian method is used. It is suitable because:

- It handles potential and kinetic energy systematically.
- It naturally includes dissipative forces via Rayleigh dissipation function.
- It provides a clean path to equations of motion without free-body diagrams.

Kinematic Model of the Shock Absorber

Only vertical motion is considered. The position of the mass is:

$$z(t)$$

The relative displacement across the shock absorber (compression) is:

$$\delta = z_{\text{mass}} - z_{\text{base}}$$

Velocity: $\dot{z}(t)$

Acceleration: $\ddot{z}(t)$

Velocity Derivation

Velocity of the mass:

$$v = \dot{z}$$

This is used to compute kinetic energy and damping power

Acceleration Derivation

Acceleration:

$$a = \ddot{z}$$

Required for Newton's second law and Lagrangian formulation.

Forces Acting on the System

Spring force (restoring):

$$F_k = -kz$$

Damping force (dissipative):

$$F_c = -c\dot{z}$$

External excitation (from road via base motion):

$$F_{\text{input}}(t)$$

Gravitational force:

$$F_g = mg \text{ (often balanced at static equilibrium)}$$

Kinetic Energy

$$T = \frac{1}{2}m\dot{z}^2$$

Potential Energy

Elastic potential energy from spring:

$$V = \frac{1}{2}kz^2$$

Gravitational potential can be omitted if z is measured from static equilibrium.

Lagrangian Formulation

Lagrangian:

$$L = T - V = \frac{1}{2}m\dot{z}^2 - \frac{1}{2}kz^2$$

Rayleigh dissipation function (for damper):

$$R = \frac{1}{2}c\dot{z}^2$$

Lagrange's equation with dissipation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{z}}\right) - \frac{\partial L}{\partial z} + \frac{\partial R}{\partial \dot{z}} = F_{\text{external}}(t)$$

Final Dynamic Equation of Motion

Substituting:

$$m\ddot{z} + c\dot{z} + kz = F_{\text{ext}}(t)$$

If base excitation $z_b(t)$ is given, relative motion $z_r = z - z_b$ leads to:

$$m\ddot{z}_r + c\dot{z}_r + kz_r = -m\ddot{z}_b(t)$$

This is the standard shock absorber equation.

Energy Dissipation Derivation

Starting from the free vibration equation (no external force):

$$m\ddot{z} + c\dot{z} + kz = 0$$

Step 1: Multiply both sides by velocity $\dot{z}$:

$$m\ddot{z}\dot{z} + c\dot{z}^2 + kz\dot{z} = 0$$

Step 2: Recognize time derivatives:

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{z}^2 \right) = m \ddot{z} \dot{z}$$

$$\frac{d}{dt} \left(\frac{1}{2} k z^2 \right) = k z \dot{z}$$

Step 3: Substitute back:

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{z}^2 \right) + \frac{d}{dt} \left(\frac{1}{2} k z^2 \right) + c \dot{z}^2 = 0$$

Step 4: Combine terms:

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{z}^2 + \frac{1}{2} k z^2 \right) = -c \dot{z}^2$$

Step 5: Final energy loss expression:

$$\boxed{\frac{dE}{dt} = -c \dot{z}^2}$$

Where total mechanical energy is:

$$E = \frac{1}{2} m \dot{z}^2 + \frac{1}{2} k z^2$$

State Space Representation

Define state vector:

$$\mathbf{x} = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}$$

Then:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} \dot{z} \\ -\frac{k}{m}z - \frac{c}{m}\dot{z} + \frac{1}{m}F_{\text{ext}}(t) \end{bmatrix}$$

In matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F_{\text{ext}}(t)$$

This is useful for simulation and control design.

Unit Consistency Check

Quantity	Unit
Displacement z	m
Velocity $\dot{z}$	m/s
Acceleration $\ddot{z}$	m/s ²
Force F	N
Stiffness k	N/m
Damping c	N.s/m
Mass m	Kg

All equations are dimensionally consistent.

Physical Meaning of Dynamic Terms

In the equation:

$$m\ddot{z} + c\dot{z} + kz = F_{\text{ext}}$$

- $m\ddot{z}$ – Inertia force (resists acceleration).
- $c\dot{z}$ – Damping force (dissipates energy, resists velocity).
- kz – Spring force (restores position).
- F_{ext} – External excitation (road input).

Conclusion

A dynamic model of a shock absorber was developed as a linear 1-DOF spring-mass-damper system. The kinematic, velocity, and acceleration relationships were defined. Using the Lagrangian method with a dissipation function, the equation of motion was derived:

$$m\ddot{z} + c\dot{z} + kz = F_{\text{ext}}(t)$$

Additionally, the energy dissipation rate was derived:

$$\frac{dE}{dt} = -c\dot{z}^2$$

The state-space representation was provided. This model forms the basis for analyzing shock absorber performance, transmissibility, ride comfort, and suspension control strategies.

VERIFICATION:

System & Equation of Motion

For a mass–spring–damper system:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

Where:

- m = mass
- c = damping coefficient
- k = spring stiffness
- x = displacement

Assume Example Values

Let:

- $m=2$ kg
- $c=4$ Ns/m
- $k=20$ N/m
- $F(t)=10\sin(2t)$

Let displacement:

$$x(t) = A \sin(2t) + B \cos(2t)$$

Step 1: Velocity

$$\dot{x}(t) = 2A \cos(2t) - 2B \sin(2t)$$

Step 2: Acceleration

$$\ddot{x}(t) = -4A \sin(2t) - 4B \cos(2t)$$

Substitute into EOM

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

Term 1: $m\ddot{x}$

$$2(-4A \sin(2t) - 4B \cos(2t)) = -8A \sin(2t) - 8B \cos(2t)$$

Term 2: $c\dot{x}$

$$4(2A \cos(2t) - 2B \sin(2t)) = 8A \cos(2t) - 8B \sin(2t)$$

Term 3: kx

$$20(A \sin(2t) + B \cos(2t)) = 20A \sin(2t) + 20B \cos(2t)$$

Combine all terms

Sine terms:

$$(-8A - 8B + 20A) \sin(2t) = (12A - 8B) \sin(2t)$$

Cosine terms:

$$(-8B + 8A + 20B) \cos(2t) = (8A + 12B) \cos(2t)$$

So LHS becomes:

$$(12A - 8B) \sin(2t) + (8A + 12B) \cos(2t)$$

Equate with RHS

Given:

$$F(t) = 10 \sin(2t)$$

Since there is **no cosine term** on RHS, the coefficient of $\cos(2t)$ must be zero.

Therefore:

$$12A - 8B = 10 \quad (1)$$

$$8A + 12B = 0 \quad (2)$$

Solve Equations

From equation (2):

$$8A + 12B = 0 \implies A = -\frac{12}{8}B = -1.5B$$

Substitute into equation (1):

$$12(-1.5B) - 8B = 10$$

$$-18B - 8B = 10$$

$$-26B = 10 \implies B = -\frac{10}{26} = -0.3846$$

Then:

$$A = -1.5 \times (-0.3846) = 0.5769$$

Final Solution

$$x(t) = 0.5769 \sin(2t) - 0.3846 \cos(2t)$$

Verification by Back – Substitution

Compute $\dot{x}(t)$:

$$\dot{x}(t) = 1.1538 \cos(2t) + 0.7692 \sin(2t)$$

Compute $\ddot{x}(t)$:

$$\ddot{x}(t) = -2.3076 \sin(2t) + 1.5384 \cos(2t)$$

Substitute into LHS of EOM:

$$m\ddot{x} = 2(-2.3076 \sin + 1.5384 \cos) = -4.6152 \sin + 3.0768 \cos$$

$$c\dot{x} = 4(1.1538 \cos + 0.7692 \sin) = 4.6152 \cos + 3.0768 \sin$$

$$kx = 20(0.5769 \sin - 0.3846 \cos) = 11.538 \sin - 7.692 \cos$$

Sum all terms:

$$\text{Sine terms: } (-4.6152 + 3.0768 + 11.538) \sin = 10 \sin$$

$$\text{Cosine terms: } (3.0768 + 4.6152 - 7.692) \cos = 0 \cos$$

$$\text{LHS} = 10 \sin(2t) = \text{RHS} \checkmark$$

Energy Dissipation Verification

Using the same values, verify $\frac{dE}{dt} = -c\dot{x}^2$

At $t = 0.2$ seconds:

$$x(0.2) = 0.5769 \sin(0.4) - 0.3846 \cos(0.4)$$

$$x(0.2) = 0.5769(0.3894) - 0.3846(0.9211) = 0.2246 - 0.3543 = -0.1297 \text{ m}$$

$$\dot{x}(0.2) = 1.1538 \cos(0.4) + 0.7692 \sin(0.4)$$

$$\dot{x}(0.2) = 1.1538(0.9211) + 0.7692(0.3894) = 1.0627 + 0.2995 = 1.3622 \text{ m/s}$$

Compute energy:

$$E = \frac{1}{2}(2)(1.3622)^2 + \frac{1}{2}(20)(-0.1297)^2$$

$$E = (1)(1.8556) + (10)(0.01682) = 1.8556 + 0.1682 = 2.0238 \text{ J}$$

Compute dissipation rate:

$$\frac{dE}{dt} = -c\dot{x}^2 = -4(1.3622)^2 = -4(1.8556) = -7.4224 \text{ W}$$

Negative sign confirms energy is being **lost** to the damper. ✓
